

Learning When to Change: Symbolic Supervision for Selective Evidence Sensitivity in Medical Reward Models

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Abstract

Reward models decide which medical reasoning traces are kept, reranked and reinforced. A useful reward reverses its preference when decisive patient evidence changes and holds it when similar-looking evidence does not apply: a penicillin allergy in the patient is a reason to change an order, the same allergy in the patient’s mother is not. We test this *selective sensitivity* with symbolic triplets from executable, stated clinical rules: a base case, a flip of one decisive input, and a near-miss that mentions the same concept without making it applicable. Labels are exact relative to the stated rule. Across 13 reward signals, 14–75% of triplets are solved: trained medical PRMs seldom reverse, and general judges reverse on near-misses. Balanced sampling teaches reversal, flip pairs alone teach over-triggering, and near-misses remove most of it. A reader that records each finding with its subject, status and time, followed by a judge with no direct access to the case, solves 79% of triplets on rules of held-out structure, 12 points more than a free-form evidence summary under the same two-stage interface. Trained on rules alone, with no medical QA data, the model raises agreement with MedEinst pair labels from 18% to 45%. Outside rules, selectivity is tested only indirectly, since existing clinical benchmarks contain no case edits of this kind.

1 Introduction

A reward model for medical reasoning is asked whether a step or an answer is right for the patient at hand. Often the answer turns on one line of the case. Under a rule that orders amoxicillin for streptococcal pharyngitis unless the patient is allergic to penicillin, the order is right for a patient with no drug allergy and wrong for one with a documented penicillin allergy (Figure 1). It is right again for a patient whose mother has that allergy. A reward that prefers the same order in all three cases fails to

Stated rule. Strep. pharyngitis: order amoxicillin; if the patient is allergic to penicillin, order azithromycin.

Claims. s : “Order amoxicillin.” s' : “Order azithromycin.”

Case	Reports	Rule	PRM	Ledger-RM
base x	allergies: none	s	+1.9	+4.2
flip x^f	allergy: penicillin	s'	+1.7	−3.9
near x^n	mother: penicillin all.	s	+1.8	+3.6
missing x^m	allergies not recorded	–	+1.6	rejects both

Figure 1: One rule, four cases. Numbers are the preference $d = u(s) - u(s')$. A PRM trained on injected step errors prefers the default order throughout; Ledger-RM reverses on the flip and holds on the near-miss.

reverse when required. One that changes its preference in the second and the third reverses under an edit that leaves the decision unchanged.

Both failures matter because reward models filter training traces (Chen et al., 2024), rerank candidates (Snell et al., 2025; Yun et al., 2025) and supply the reward in reinforcement learning (Shao et al., 2024; Guo et al., 2025). The first is documented: LLM judges are more reliably invariant to irrelevant edits than sensitive to relevant ones (Chen et al., 2026b), and clinical LLMs mention a changed patient fact in 85.6% of their rationales while keeping the original medication choice in 69.8% (Hou et al., 2026). The second is easy to overlook: an output change after a targeted edit means little unless it exceeds the change after an edit that should not matter (Yang et al., 2026). Medical reward models are evaluated on answer selection and on injected step errors (Yun et al., 2025; Wu et al., 2026), neither of which changes the case. Formal supervision transfers to informal tasks (Kamoi et al., 2026); which interventions teach a reward when evidence applies is open.

We build the test from executable rules (§3). Each rule yields *triplets*: a base case, a *flip* of one decisive input, and a *near-miss* in which the decisive concept appears without counting under the rule: in a relative, negated, at a time the rule does

not count, or as a value that approaches a threshold without crossing it. Each case is crossed with the two claims the rule can support, and the rule program labels all six instances. Labels state what a stated rule implies, not what is clinically best. Clinical benchmarks with existing labels form a second tier, where each measures a different construct (Table 1). Our contributions:

(1) **A test.** Triplets separate two failures that pair tests conflate. Across 13 reward signals 13.8–75.0% of triplets are solved (§7.1).

(2) **A controlled comparison** of training distributions and intermediate representations under one interface and budget (§7.2): balance teaches reversal, flip pairs alone teach over-triggering, near-misses correct it, and a typed applicability ledger adds 12.1 points over a free-form summary read by the same case-blind judge. On rules the gain lies in what the reader is trained to record: a judge given one applicability bit does almost as well.

(3) **Transfer with its confounds removed.** A model trained on rule triplets only reaches 78.6% on rules of held-out structure and raises agreement with MedEinst labels from 18.4% to 44.8% without medical QA data (§7.3).

2 Related Work

Validity of evaluators. Chen et al. (2026b) define a judge’s validity as invariance to construct-preserving edits together with sensitivity to construct-changing ones. Yang et al. (2026) show that the effect of a targeted counterfactual edit must be compared with that of meaning-preserving edits of the same size; our near-miss and presentation edits are such baselines with known labels. Robustness work on reward models targets response-side artifacts (Liu et al., 2025; Srivastava et al., 2026; Zang et al., 2025; Xu et al., 2025). The closest is DynaCF, which rewrites the chosen response without changing its content and down-weights training pairs whose margin shifts or flips (Liu et al., 2026); its edits should not matter and serve as probes, whereas we edit the case in both directions and train on the edits, whose labels are exact. CSR regularises a solver to change its answer when an operator in its own trace is swapped (Shihab et al., 2025).

Process rewards and symbolic supervision. PRMs are trained on human step labels (Lightman et al., 2024), rollout estimates (Wang et al., 2024) or generated verification reasoning (Khalifa

et al., 2025), which GenPRM extends with executed code (Zhao et al., 2025). FoVer trains step verifiers on labels from formal tools and shows transfer to informal tasks including NLI (Kamoi et al., 2026), so transfer of formal supervision is not our claim; we ask which interventions teach applicability and compare with a model trained on FoVer’s data. Other work perturbs steps and keeps a verifier at inference (Zi et al., 2026), injects errors into verified chains (Chi and Wang, 2026), or uses a deterministic verifier as the reward (Pronesti et al., 2026). In medicine, Med-PRM verifies steps against retrieved guidelines (Yun et al., 2025), MedCEG rewards a policy for covering a reference evidence graph built for each training question (Mu et al., 2025), and MedPRMBench injects 14 error types into chains from seven QA sources (Wu et al., 2026). MedGuideX compiles guideline recommendations into executable functions and post-trains a clinical policy on factual and counterfactual questions generated from them, keeping only interventions that change the outcome (Shen et al., 2026); executable guideline functions are also called at inference (Cao et al., 2026). None of these evaluates a reward model on a changed case. Mediating a reward through concepts (Koh et al., 2020; Laguna et al., 2025) precedes our ledger, as does EVPV, which extracts visual facts once per problem, independently of the solution, and attenuates the reward of steps whose stated premises they do not support (Wang et al., 2026). The fields of our ledger are the attributes that clinical NLP assigns to mentions of findings: negation, experiencer and temporality (Chapman et al., 2001; Harkema et al., 2009; Uzuner et al., 2011). Typed evidence records and distractors that are similar but irrelevant have been studied in retrieval (Zhou et al., 2026; Agand, 2026; Chen et al., 2026a), and conditional QA asks for answers that change with a stated patient condition (Parekh et al., 2026). The space left open is the reward model: a test with near-misses whose labels are exact, their use as supervision, and a judge that sees only the record.

Context against prior, and counterfactual data.

How language models arbitrate between memory and conflicting context is well studied (Xu et al., 2024; Wu et al., 2024; Shi et al., 2024). Clinical policies keep the prototypical answer when decisive evidence changes (Chen et al., 2026c; Hou et al., 2026; Kim et al., 2025; Qu and Färber, 2026), and clinical-trial NLI has been probed with label-

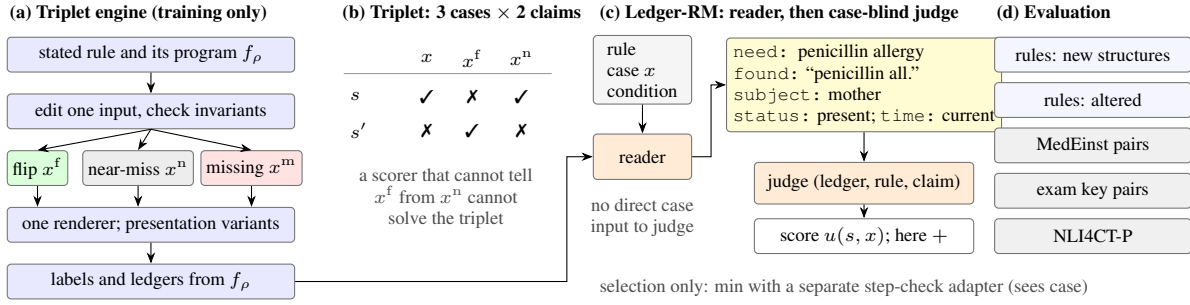


Figure 2: Overview. The rule program (blue) supervises reader and judge and is not executed at inference. The reader records findings with subject, status and time; the judge scores the claim from the ledger and the rule, with no direct case input. A string check on quotations is the only symbolic operation at inference.

preserving and label-inverting interventions (Julien et al., 2024). Minimal label-flipping edits serve training (Kaushik et al., 2020) and evaluation (Gardner et al., 2020; Ribeiro et al., 2020; Thrush et al., 2022), with the caveat that narrow edits may not help out of distribution (Joshi and He, 2022); claim-only scorers are partial-input baselines (Polia et al., 2018).

3 Triplets from Two Tiers

3.1 Rule tier

Rules, states, claims. A rule ρ has criteria $c_1, \dots, c_m$ over typed inputs. A patient state z is a set of *mentions*; each has a concept, a value, a subject (patient or other), a status (present, absent, unknown) and a time (current or past). Every criterion c_j comes with an *applicability predicate* A_j that says which mentions it counts. The predicate is part of the rule: a creatinine threshold counts the patient’s current value, a stroke-risk score counts a stroke at any time, and some risk scores count disease in a first-degree relative. In 23% of criteria a past or family finding counts, so no global heuristic about time or subject is correct. *Scoring rules* are clinical calculators, with claims of the form “criterion j adds b points”. *Constraint rules* state a decision in full: choose a unless condition c holds, then b ; the claims are the two options in neutral wording of matched length. The prompt states the rule. A contraindication can exclude an order, and no single rule establishes that an alternative is optimal; our labels claim only the former kind of fact.

Edits and invariants. The engine edits the mentions of one input of a criterion c_j and keeps an edit only if executing the program confirms its invariant. A *flip* changes a mention that A_j counts

so that c_j changes and no other criterion does. A *near-miss* leaves every criterion unchanged; it is *numeric* (a counted value moves toward the threshold without crossing) or concerns *applicability* (the concept and value of the flip are reported in a mention that A_j does not count: another subject, absent status, or a time outside A_j). A *presentation* edit re-renders the same state with another unit, order or template, matched in size to the flips (Yang et al., 2026). For *missing* twins we distinguish explicit absence, which determines a criterion, from an unknown input. Measured inputs are unknown when omitted; history items are absent when omitted, as in the source benchmarks, and unknown only when the case says so (“allergies not recorded”). With $C_\rho(z')$ the completions of a partial state z' within the input ranges, a missing twin is kept only if $|\{c_j(z) : z \in C_\rho(z')\}| > 1$. A verdict “+” means the claim is supported by the case under the rule; “−” covers refuted and undetermined claims, which verifiers for retrieval keep apart (Qiu et al., 2026; Kamelhar, 2026) and we report separately.

Triplets. A triplet crosses a base case x , its flip x^f and a near-miss x^n with the claims s (correct for x and x^n) and s' (correct for x^f). For a reward with score $u(s, x)$ let $d(x) = u(s, x) - u(s', x)$. The triplet is *solved* if $d(x) > 0$, $d(x^f) < 0$ and $d(x^n) > 0$; ties are failures.

Proposition 1. *Let u be deterministic. (i) If $u(s, x) = g(s)$, then $d(x) = d(x^f)$. (ii) If $u(s, x) = g(x)$, then $d \equiv 0$. (iii) If $u(s, x) = g(s, h(x))$ for a feature map h with $h(x^f) = h(x^n)$, then $d(x^f) = d(x^n)$. In each case the triplet is not solved; under (iii) a correct reversal on the flip implies failure on the near-miss.*

Proof. (i) and (ii) are immediate from the definition of d . (iii) $d(x^f) = g(s, h(x^f)) - g(s', h(x^f)) =$

Source	Label basis	What it can show
Rule triplets	executed program of a stated rule	reversal and holding under that rule; exact
MedEinst	DDXPlus cases, edited and filtered by the benchmark	agreement with labels across paired diagnostic cases
Key pairs	answer keys of two exam questions	paired answer discrimination
NLI4CT-P	expert labels, controlled interventions	response to statement edits, report fixed

Table 1: Sources and constructs. Only the first contains near-misses.

$g(s, h(x^n)) - g(s', h(x^n)) = d(x^n)$, while a solved triplet needs $d(x^f) < 0 < d(x^n)$. $\square$

Let $h_k(x) \in \{0, 1\}$ indicate whether the decisive concept k is named anywhere in the case, whatever its value, subject, status or time. For a finding that the flip introduces, the base case is rendered without naming k and an applicability near-miss names it, so $h_k(x) = 0$ and $h_k(x^f) = h_k(x^n) = 1$: (iii) applies. For a measured input k is named in all three cases, so $d(x) = d(x^f)$ as in (i). A scorer that uses the case only through h_k thus solves no triplet, although it can reverse on every flip of the first kind. Richer attribute-blind features need not collide and are an empirical baseline (Appendix F). Flip pairs alone exclude only (i) and (ii). With sampled scores the statement holds in expectation, and finite samples can show apparent reversals; we report tie rates and sampling intervals.

Metrics. *Reversal* (Rev): share of base–flip pairs with $d(x) > 0$ and $d(x^f) < 0$. *Hold*: share of near-misses with $d(x^n) > 0$. *Triplet accuracy* (TA): share of solved triplets, the primary summary. None needs a threshold. Base accuracy, ties, and the conditional rate of unnecessary reversal with its denominator are in Appendix F. For missing twins, MR is the rate at which both claims are rejected at the threshold that gives 5% false rejection of supported claims on development data.

3.2 Clinical tier

Labels that we generate with one model and approve with another would not be ground truth, so the clinical tier uses only released labels. Each source supports a different statement (Table 1).

MedEinst (Chen et al., 2026c) pairs a control case with a trap case that differs in discriminating findings and has another diagnosis (5,383 test pairs; 10,689 reference pairs). *Key pairs* are two exam questions whose keyed answers each occur

among the other’s options. A claim is the complete question with one candidate answer, so the four labels follow from the keys. We exclude negatively phrased questions and options that refer to other options, and keep each question in at most one pair. Two exam questions are not a controlled counterfactual; the test measures discrimination between answers given the case. We mine pairs from MedQA (Jin et al., 2021) and CareQA (Arias-Duart et al., 2025). *NLI4CT-P* (Jullien et al., 2024) edits expert-written statements about trial reports; we report its faithfulness and consistency with base F1.

4 Diagnostics

Parts and whole. For each base–flip pair we build a *reading* pair, whose claims quote the decisive finding, and an *application* pair, which scores the original claims against a one-line case with only that finding. With P, K, U the events of a correct reversal on the three pairs, the composition gap is $G = 1 - \Pr[U \mid P \wedge K]$; it is descriptive, since the tasks differ in length and in what must be selected.

Appendix F analyses how far, and in which direction, a decisive edit moves the preference relative to a missing-input baseline.

5 Ledger-RM

Reader. The reader R_ϕ receives the case x , the rule ρ where one is stated, and the *condition under test* ω : the criterion, or the set of candidate answers, on which the claims disagree. It does not receive a candidate’s asserted outcome, so the claims of a triplet are judged from one ledger $e = R_\phi(x, \rho, \omega)$ of up to 3 entries. An entry records facts and no decision: the condition the claim depends on (need); what the case says about it (found: a value with unit, a verbatim quotation, or missing); whose finding it is (subject); its status (present, absent, unknown); and its time (current or past). Keys and categorical fields are fixed by a grammar, and a quotation that does not occur in the case makes the ledger malformed.

Judge. The judge J_ϕ receives the ledger, the rule if any, and the claim, and decides whether the recorded finding counts under the rule and what follows for the claim. The end-to-end score is

$$u(s, x) = J_\phi(R_\phi(x, \rho, \omega), \rho, s), \quad (1)$$

with $J_\phi(e, \rho, s) = \log p_\phi(+ \mid e, \rho, s) - \log p_\phi(- \mid e, \rho, s)$. The judge has no direct case input: the

case reaches the verdict only through the ledger. A malformed ledger receives the floor score $-L$ ($L = 10$), so two malformed ledgers give a tie, which counts as a failure. Reader and judge are two prompts of one adapter.

Training. On rules the program supplies the target ledger $e^*(x, \rho, \omega)$ and, for any ledger e , the verdict $y_\rho(e, s)$ obtained by executing the rule on the state that e describes. Let $N(x)$ be the other training cases of the same rule and condition, including the twins of x . The judge is trained on a ledger $e \sim q(\cdot \mid x, \rho, \omega)$, which equals $e^*(x, \rho, \omega)$ with probability $1 - p$ and $e^*(x', \rho, \omega)$ for x' drawn uniformly from $N(x)$ otherwise:

$$\mathcal{L} = \mathbb{E}_{(x, \rho, s)} \left[-\frac{1}{|e^*|} \log p_\phi(e^* \mid x, \rho, \omega) - \lambda \mathbb{E}_{e \sim q} \log p_\phi(y_\rho(e, s) \mid e, \rho, s) \right]. \quad (2)$$

A resampled ledger is a valid input for the judge, whose target is the verdict that ledger implies; no instance rewards a verdict that contradicts the evidence shown to the judge.

What follows by construction, and what does not. The verdict is computed from the ledger. That the ledger is correct and sufficient, and that the judge uses its fields as intended, is learned; we test the latter by field interventions and by ablations that add or isolate a decision field (§7.4). A missing or misattributed entry cannot be repaired by the judge; we measure the share of errors this causes with program-supplied ledgers and test a second pass that re-reads every entry against the case (Agand, 2026).

Step check and combined reward. Errors internal to a step are scored by a separate adapter, the injected-error PRM, which sees the case. A trace is acceptable only if both checks pass, and whatever the dependence between the two events, their joint probability is at most the smaller of the two. For candidate selection we therefore rank by the minimum of the two scores after temperature scaling on development data; the product, a combination fitted on development data and reliability gating (Wang et al., 2026) are compared in Appendix G. In selection a step is scored alone, so the reader receives the step and the ledger can depend on it (Appendix E). All other results use the ledger score alone, and the ledger adapter is never trained on step-error or QA data unless a row says so.

6 Experimental Setup

Rules and shifts. The library has 48 MedCalc-Bench calculators (Khandekar et al., 2024) that two audits do not flag (Krohn-Grimberghe, 2026; Ye et al., 2025), 90 further published scores that we implemented from their definitions, 120 constraint rules over renal function, pregnancy, allergy, age and interacting drugs, compiled from public recommendations by the tree-then-program procedure of Shen et al. (2026), and 300 rules sampled from a typed grammar over clinical inputs. Every program passes boundary, unit and dependency tests against the rule text shown in the prompt. Each rule has a *structural signature* (rule kind, operators, dependency depth, input types, applicability predicate). We test on trained rules with new patients (L0), unseen rules of seen signature (L1), held-out signature classes (L2), and trained rules whose thresholds are changed in the prompt, restricted to cases on which the original and the altered rule disagree (L3-alt). Test cases use rendering templates and cue phrases for relatives, negation and time that do not occur in training (overlap statistics in Appendix B).

Training distributions. All trained models use Qwen3-8B (Yang et al., 2025), LoRA (Hu et al., 2022) and 160k instances. Four corpora share one instance format and the same share of missing-input and presentation instances (15% each) and differ in one ingredient at a time. *Natural*: empirical patient states, each case with the correct and the counterfactual claim; a simplified stand-in for step-perturbation corpora (Zi et al., 2026; Chi and Wang, 2026), which we do not reimplement. *Balanced*: each criterion outcome equally frequent. *Blocks*: the balanced distribution with cases in base-flip pairs. *Triplets*: blocks in which near-misses replace an equal number of base cases.

Representations. Two one-stage formats, in which the verdict is produced with the case in context: verdict only, and a rationale followed by the verdict. Three two-stage formats with the same case-blind judge interface and the same token budget: a free-form evidence summary, a value ledger (need, found), and the applicability ledger.

Systems. *Audited signals*: Med-PRM (Yun et al., 2025), the MedS³ PRM (Jiang et al., 2025), the FoVer PRM (Kamoi et al., 2026), a PRM trained on the MedPRMBench training split (injected-error PRM), ThinkPRM (Khalifa et al., 2025) and GenPRM-7B with code execution (Zhao et al.,

2025); Qwen3-8B, Qwen3-32B, Llama-3.3-70B and Kimi-K2; GPT-5.4, Gemini 3.1 Pro and Claude Opus 5.5. *Trained comparisons* on Qwen3-8B: FoVer’s training data; the five representations; and a GenPRM-style verifier, trained on the same triplets to write an analysis and a Python check of the claim, whose executed output precedes the verdict (targets rendered from the rule program). MedCEG’s reward needs a reference graph for the question, so it cannot score a claim alone; we use one of that kind only in policy training (Appendix G). *Training-free*: default correction, $u(s, x) + \alpha [u(s, x) - u(s, \emptyset)]$ (Shi et al., 2024), which amplifies the contribution of the case; a prompted ledger; and a verifier that writes and executes a program for the stated rule (Ocampo Osorio et al., 2026). *References*: the best closed judge with the prompted ledger; extraction followed by a hand-written program.

Clinical data and protocol. MedEinst test pairs, 1,240 key pairs from MedQA test and 960 from CareQA, and NLI4CT-P; versions are pinned (Appendix D). Scores are verdict log-odds where token log-probabilities are available and verdict frequencies over 32 samples otherwise. We train 5 seeds. Intervals come from a bootstrap over signature classes and rules, diseases, or questions, with triplets kept together and seeds included. Primary endpoints are triplet accuracy on L2 and MedEinst reversal of the rule-only model, with 5 Holm-corrected comparisons; systems and thresholds are selected on development splits and frozen before test (Appendix A).

7 Results

7.1 Audit of existing reward signals

The two failures separate the signals (Table 2). Trained PRMs seldom reverse (14.9–33.5%) and therefore hold; they solve 13.8–25.6% of triplets. Open judges reverse more often and hold on only 66–72% of near-misses. Closed judges solve 71.0–75.0%. Reversal is lower on MedEinst for every signal and higher on key pairs, which are not minimal. Shortcut scorers show what each test excludes (Appendix F): a classifier that sees only the step reaches a macro-F1 of 81.7 on the MedPRMBench test split, against 41.2 for the majority class, and solves no triplet; a scorer that sees only whether the decisive concept is named solves no triplet, although it reverses on every flip that introduces the concept; a logistic model on attribute-blind

Reward signal	Rule triplets				ME	Key
	Rev	Hold	TA	MR	Rev	Rev
<i>Trained PRMs</i>						
MedS ³ PRM	14.9	91.0	13.8	9.5	12.1	33.0
Med-PRM	19.6	89.5	17.7	12.8	15.3	40.2
FoVer PRM	22.4	87.2	19.8	16.0	11.0	24.7
Inj.-error PRM	23.8	90.1	21.5	15.2	13.6	36.9
ThinkPRM	30.2	71.5	22.4	20.3	14.8	38.5
GenPRM-7B	33.5	73.0	25.6	22.0	15.5	39.0
<i>Open judges</i>						
Qwen3-8B	37.5	66.3	27.9	24.1	18.4	44.5
Llama-3.3-70B	58.9	68.0	42.1	41.5	33.0	60.7
Qwen3-32B	62.0	70.2	45.3	44.9	35.8	63.2
Kimi-K2	68.7	72.4	51.0	50.6	41.2	68.0
<i>Closed judges</i>						
GPT-5.4	84.1	83.0	71.0	66.2	55.6	79.9
Gemini 3.1 Pro	85.0	84.1	72.6	67.0	57.1	81.0
Claude Opus 5.5	86.3	85.9	75.0	68.0	58.3	82.4

Table 2: Audit (%), each signal with its native readout (Appendix F). Rule triplets, L0: reversal, hold, triplet accuracy, missing-evidence rejection at 5% false rejection. ME, Key: reversal on MedEinst and on key pairs.

Representation	Nat.	Bal.	Blocks	Triplets
<i>One-stage (verdict produced with the case in context)</i>				
Verdict only	9.8	24.0	31.5	52.4±2.1
Rationale, then verdict	14.1	29.7	37.2	61.0±1.9
<i>Two-stage (case-blind judge)</i>				
Evidence summary	16.0	31.8	39.6	66.5±1.8
Value ledger	17.9	33.4	41.8	68.3±1.7
Applicability ledger	20.6	36.1	44.0	78.6±1.6

Table 3: Triplet accuracy on L2 (%) by training corpus and representation; ±: s.d. over 5 seeds. Qwen3-8B, rule data only, equal instances and steps; rows other than the first use the same token budget. Corpora differ in one ingredient at a time (§6).

concept–value features solves 27.2%, almost all of them numeric triplets.

Every signal reverses correctly on at least 84.9% of reading and of application pairs and fails the full pair in 11–80% of the cases in which it solves both (Appendix F).

7.2 Training distribution and representation

Balance teaches reversal. From the natural to the balanced corpus a verdict-only model rises from 9.8% to 24.0% triplet accuracy (paired difference 14.2, interval [10.1, 18.6]) and its reversal from 21.3% to 71.8% (Table 3).

Flip pairs alone teach over-triggering. Trained on blocks, the verdict-only model reverses on 78.9% of flips and holds on 39.8% of near-misses. Replacing base cases by near-misses raises hold to 70.1% at a reversal of 76.5% and triplet accu-

racy from 31.5% to 52.4% [+16.4, +25.5]. Flip-only data is what a pipeline yields when it keeps only interventions that change the outcome (Shen et al., 2026) or rewards change under an edit (Shihab et al., 2025); both leave over-triggering unconstrained. Using near-misses only as probes for re-weighting, as DynaCF does with surface rewrites, recovers part of the gain (Table 9).

Applicability structure adds to a matched two-stage baseline. Under one judge interface and budget, the applicability ledger solves 78.6% of triplets, the value ledger 68.3% and a free-form summary 66.5%; the difference to the summary is 12.1 points [8.0, 16.4]. The second stage accounts for 5.5 points (summary against one-stage rationale). Without near-misses in training the fields buy little (44.0% against 41.8%).

7.3 Transfer within and beyond rules

Within rules. Trained on rule triplets only, Ledger-RM solves 78.6% of triplets on rules of held-out structure and 69.0% on altered rules where the original rule gives the other answer; the untrained critic solves 11.0% of the latter (Table 4). Training on FoVer’s formal data does not produce this behaviour (27.3%). Default correction raises reversal and lowers hold from 66.0% to 52.5%.

From rules to clinical labels. The rule-only model has seen no medical QA data of any kind. Its agreement with MedEinst pair labels rises from 18.4% to 44.8% [39.9, 49.6], against 39.4% for the free-form summary. Adding MedPRMBench step-error data to the adapter adds 3.8 points, which is medical supervision and not transfer. Near-misses buy holding at a small price in reversal: against the blocks model, the triplets model loses 1.5 points on MedEinst and 1.4 of NLI4CT-P faithfulness and gains 6.6 of consistency. Trained on clinical pairs, the model reaches 66.8% on MedEinst; this is in-distribution, and with diseases held out it is 58.9% (Appendix D).

Alternatives. A GenPRM-style verifier trained on the same triplets writes and executes a check for every claim. It reaches 74.0% on L2 and is the most robust trained model on altered rules (72.5%), at 11 times the generated tokens; on MedEinst, where there is nothing to execute, it is at 35.8%. Executed code settles how a rule is applied and not how the case is read: in 81% of its near-miss errors a relative’s or a past finding is passed to the code

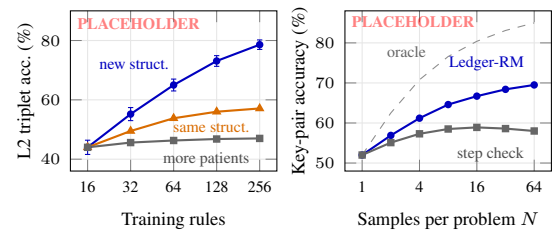


Figure 3: Left: L2 triplet accuracy of Ledger-RM as training grows by adding rules of new structures, rules of structures already seen, or only patients for 16 fixed rules (same data size; bars: s.d. over seeds). Right: accuracy on both members of a MedQA key pair for the trace selected among N samples.

as the patient’s. The closed judge with a ledger prompt leads on L2, key pairs and NLI4CT-P.

7.4 What the judge uses

A decision bit explains most of the rule-tier gain. If the reader also writes whether the condition applies, a judge that sees only that bit and the claim reaches 77.5% on L2, against 78.6% for the fact-only ledger. On rules the benefit therefore lies in what the reader is trained to establish, not in the judge’s use of the fields. The fields matter in transfer: 38.2% against 44.8% on MedEinst zero-shot (Table 9). They also matter as supervision: a reader trained to write the bit without fields reaches 63.0%.

The reader is the bottleneck. With program-supplied ledgers the judge solves 96.5% of L2 triplets, so about 84% of the errors of the full model arise in the reader. A second pass that re-reads each entry against the case and regenerates rejected ones raises triplet accuracy to 81.3%.

Direct access to the case trades selectivity for coverage. A one-stage variant, in which the ledger is written as a rationale and the verdict is produced with the case in context, solves 71.6% of L2 triplets, holds on 80.2% of near-misses against 86.9%, and is 2.7 points better on MedEinst zero-shot. Where a few typed fields cannot carry the evidence, the bottleneck costs accuracy.

Field interventions. Changing subject, status or time in a correct ledger changes the verdict as the program prescribes in 97.3% of cases, and given the ledger of another case the judge returns the verdict that ledger implies in 96.8% (Appendix E). Figure 3 (left): rules of new signature classes raise L2 accuracy from 44.0% to 78.6%, rules of seen classes to 57.1%, more patients to 47.0%.

System	Rule tier				Clinical tier			
	L2	L3-alt	Hold	MR	MedEinst	Key pairs	NLI-F	NLI-C
<i>Training-free, Qwen3-8B</i>								
Critic	25.1	11.0	66.0	24.1	18.4	44.5	60.2	83.0
+ default correction	30.4	17.9	52.5	24.6	27.0	49.3	64.9	80.2
+ prompted ledger	39.2	24.3	70.1	46.3	26.1	48.0	66.1	82.4
Generated-program verifier	68.5	63.7	88.4	58.4	—	—	—	—
<i>Trained without medical QA data (clinical columns are zero-shot)</i>								
Verdict only, FoVer data	27.3	15.8	69.0	22.0	19.9	43.8	63.5	84.0
Verdict only, rule triplets	52.4	34.0	70.1	77.2	33.0	49.0	65.2	84.1
Evidence summary, rule triplets	66.5	50.3	79.0	80.4	39.4	51.6	68.0	85.2
GenPRM-style verifier, rule triplets	74.0	72.5	83.0	71.5	35.8	50.2	66.9	84.8
Ledger-RM, rule balanced	36.1	27.5	60.3	81.0	37.2	50.1	67.4	82.6
Ledger-RM, rule blocks	44.0	35.2	49.0	83.9	46.3	53.5	71.2	79.4
Ledger-RM, rule triplets	78.6$\pm$1.6	69.0	86.9	84.6	44.8 $\pm$ 2.3	52.9	69.8	86.0
<i>With medical data</i>								
Ledger-RM, rule triplets + step-error data	78.0	68.1	86.2	84.3	48.6	55.3	71.5	86.1
Ledger-RM, clinical pairs only	29.8	16.9	62.0	41.0	62.4	72.9	80.7	87.5
Ledger-RM, rule triplets + clinical pairs	77.9	68.2	86.4	84.0	66.8	75.6	83.9	88.2
<i>References</i>								
Claude Opus 5.5, zero-shot	73.4	43.9	85.5	68.0	58.3	82.4	87.2	90.3
Claude Opus 5.5, prompted ledger	85.1	70.6	91.2	81.9	63.0	84.1	89.5	90.8
Extraction + hand-written program	96.0	96.2	98.0	94.0	—	—	—	—

Table 4: Transfer (%), ledger score alone, 5 seeds. L2, L3-alt: triplet accuracy on rules of held-out structure and on altered rules; Hold, MR: on L2; MedEinst, key pairs: reversal; NLI-F, NLI-C: faithfulness and consistency on NLI4CT-P. Clinical pairs: MedEinst reference pairs and MedQA-train key pairs, so MedEinst and key pairs are in-distribution for those rows. The two verifier rows execute code written by the model; the last row executes a hand-written program on an extracted state, and its gap to 100 is extraction error. Bold: best among models trained without medical QA data.

Selection	Accuracy		Key pair	MedEinst		
	MQA	CQA		pair	ctrl	trap
Single sample	71.2	66.8	52.0	31.2	78.5	39.8
Self-consistency	74.5	70.1	55.8	33.5	81.0	41.4
Step check	77.6	72.4	58.9	34.0	81.2	41.9
vignette swapped	76.8	71.9	57.6	33.6	80.9	41.5
Ledger score	76.0	71.0	65.1	44.7	81.3	54.9
Combined	78.3	73.0	66.7	45.9	82.0	55.6
vignette swapped	73.9	69.2	54.1	32.8	80.1	40.6
no near-misses	77.1	72.2	64.9	46.5	81.0	56.8
Claude Opus 5.5	79.5	76.0	70.2	44.1	82.0	53.8
Oracle selection	88.0	85.3	80.4	58.7	90.2	65.1

Table 5: Reranking 16 samples of a Qwen3-8B policy on identical pools (%). MQA: MedQA; CQA: CareQA. Step check: injected-error PRM. Ledger score: Ledger-RM trained on rule triplets and clinical pairs. Combined: minimum of the two after calibration.

7.5 Candidate selection

Table 5 tests selection among samples of a fixed policy. The combined reward and the step check improve accuracy on MedQA and CareQA by similar amounts (difference 0.7 [−0.2, 1.6] on MedQA, non-inferior at a margin of one point). On both members of a key pair the combined reward reaches 66.7% against 58.9%, and on MedEinst pairs 45.9%

against 34.0%, with control accuracy unchanged. Swapping in the vignette of another question removes the gain of the combined reward and leaves the step check almost unchanged: the ledger score depends on the supplied case. That the dependence is correct is what the pair columns show. Without near-misses in training the combined reward loses 1.2 points on MedQA and nothing on MedEinst, whose pairs contain no near-misses. As N grows, pair accuracy under the step check stays flat while the oracle rises (Figure 3, right).

8 Conclusion

On executable rules, reward models fail in two directions. The remedy has two parts: balanced training with near-misses, and a record of what the case says, whose it is and when, from which a case-blind judge decides. Trained on rules alone, this carries to rules of unseen structure and, in part, to existing clinical pair labels. Holding on inapplicable evidence in clinical text is not established.

Limitations

What the labels are. Rule-tier labels are exact relative to a stated rule, its applicability predi-

cates and a rendering grammar. They say what the rule implies, not what is clinically optimal. We commissioned no new clinical annotation; reused benchmarks carry the review their authors report. Clinical-tier results are agreement with released labels: MedEinst cases derive from a simulator and were edited and filtered by the benchmark, key pairs inherit the errors of exam keys and are not counterfactuals, NLI4CT-P edits statements about breast-cancer trial reports. Nothing here supports clinical use.

Selectivity outside rules is tested only indirectly. Applicability near-misses with exact labels require a rule program. On the clinical tier, holding is tested only on the claim side: by NLI4CT-P consistency under statement edits and by expert-reviewed distractors in real discharge summaries that attribute a correct fact to the wrong admission (Kim et al., 2026) (Appendix D). No existing benchmark edits the case in this way; assertion annotations of clinical notes (Uzuner et al., 2011) would allow a test of the reader alone.

The bottleneck. The ledger assumes that a claim depends on a few identifiable findings with a subject, status and time. Claims that weigh many soft findings fit it poorly, and the one-stage variant is better on MedEinst. The judge computes its verdict from the ledger; nothing guarantees that the ledger is right, and a quotation can exist in the case and still be the wrong evidence.

Rule population. Rules come from one calculator benchmark, our implementations of published scores, stated constraints and a grammar; templates are cleaner than clinical notes. Held-out signature classes share operators with training rules.

Weak supervision. For rows trained on clinical pairs, ledger targets come from differences between paired cases and from selection with a frozen judge; they are not verified.

Interpretation and scope. The composition gap and shift classes are descriptive. Reranking tests selection for one policy. All data are in English.

Ethical Considerations

The work uses public benchmarks and synthetic patients; no patient data were collected. Stated rules in our data are test specifications and must not be read as prescribing guidance. A reward model that appears to check patient evidence could be trusted more than it should be; accuracy on these triplets does not establish safety in care, and the

models are released for research on post-training only. Because the engine generates cases on which a given reward fails, it could be used to construct inputs that mislead a deployed evaluator; we see this as an argument for auditing rewards before they are used.

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A Analysis Plan

Draft status. No experiment in this manuscript has been run. Every value printed in red, and every curve marked as placeholder data, is a target value used to lay out the paper. This paragraph and the red banner disappear, and the build fails on any remaining placeholder, once measured values are entered.

Primary endpoints. Triplet accuracy on L2 (rendered text, rule stated) and reversal on MedEinst test pairs for the model trained on rule triplets only.

Primary comparisons (paired, Holm-corrected): on L2, (1) balanced against natural, verdict only; (2) triplets against blocks, verdict only; (3) applicability ledger against two-stage evidence summary, triplets; on MedEinst, (4) rule-only Ledger-RM against the untrained critic; (5) rule-only Ledger-RM against the rule-only evidence summary.

Decision rules. If (2) fails, near-misses are reported as an evaluation device only. If (3) fails, the paper is a data-design result. If (4) fails, the answer outside rules is negative. If the gain in (4) appears only with step-error data, it is reported as a gain from medical supervision.

Selection and freezing. Hyperparameters, thresholds, the calibration of the combined reward and the choice of headline system use development splits only: development triplets from training-fold signature classes and a held-out part of the MedEinst reference set. Test sets are generated or opened after these choices are frozen and are evaluated once.

B Rule Library and Splits

Sources. We use the calculator programs released with MedCalc-Bench. Further scores are implemented from their published definitions; the manifest gives the source of each, the rule text shown in prompts, the program and its tests. We do not use the calculator set of MedCalc-Eval (Mao et al., 2025), whose definitions were not available to us.

Source	Rules	Classes
MedCalc-Bench calculators (audited)	48	9
Published scores, our implementation	90	11
Stated constraint rules	120	5
Grammar-sampled rules	300	19

Table 6: Rule population. Classes overlap across sources; 19 in total.

Because every prompt states the rule, correctness is relative to the stated text for all four sources.

Constraint rules and claims. Each rule states a default option, a condition and an alternative. It is derived from a public drug-label or guideline recommendation in three steps taken from Shen et al. (2026), whose own functions are not released: a model extracts the recommendation, writes it as a decision tree, and compiles the tree to a program. We keep a rule if the tree is complete, the program passes the boundary tests and agrees with the tree on sampled inputs, and show the verbalised tree in the prompt. No clinician checks the tree against the source, which is why labels are relative to the stated rule. Options are plain orders of one form (drug, dose, route, frequency, duration) that differ by at most 2 tokens and contain no word that gives a reason. Every case reports the same fields with raw values, so the presence of a field carries no label information. Swapping which option is the default, or paraphrasing the options, changes triplet accuracy of Ledger-RM by at most 0.9 points.

Splits and overlap. In each of 3 folds 5 of 19 signature classes are held out for L2. Between training and L2 rules, 0% of canonical programs, 12.4% of criterion subexpressions, 0% of rule texts and 0% of rendering templates and cue phrases are shared: each input type has 6 templates, 4 for training and 2 for test, and the words that mark relatives, negation and time are split in the same way. With training templates and cues on the same L2 rules, triplet accuracy is 81.4% for Ledger-RM, against 78.6% with held-out ones. L3-alt moves one or two thresholds and keeps only cases whose outcome differs between the original and the altered rule.

C Triplet Construction

Invariants. Of the proposed flips, 7.9% are discarded because a second criterion changes; of the proposed missing twins, 18.6% because the remaining inputs determine the criterion. Near-misses are checked by executing the program on the state re-

covered from the rendered case, and the equalities $h_k(x) = 0$, $h_k(x^f) = h_k(x^n) = 1$ (or $h_k \equiv 1$ for measured inputs) are checked on the rendered text of every triplet. Per fold the training set has 16,000 triplets over 370 rules and L2 has 8,000 over 130. The authors inspected 300 rendered triplets, stratified by edit kind, for rendering errors and found 2; this checks the grammar and not clinical validity.

Near-miss kinds. Numeric 40%, other subject 20%, absent status 20%, time outside the predicate 20%. For criteria whose predicate counts past or family findings, such mentions are flips.

Text tiers. Notes rewritten by Gemma-3-27B are accepted if two extractors recover the intended state (8.7% rejected). Ledger-RM reaches 74.8% on accepted and 60.3% on rejected groups relabelled from the intended state, so the filter removes harder text.

D Clinical Tier

Releases. MedEinst as released by its authors; NLI4CT-P as distributed for SemEval-2024 Task 2; MedPRMBench and CareQA at pinned revisions listed in the repository.

MedEinst. Claims are the two diagnoses. For training rows, the found targets are the findings in which the two cases differ, taken from the structured evidence where released and from a token diff otherwise; the two agree on 96.2% of a sample of 500 pairs, and in 97 of 100 pairs inspected by the authors the difference names the edited finding. Other ledger fields are not supervised there. In the disease-held-out split a pair is in the test set only if both of its diagnoses are test diseases (12 of 49), and in the training set only if both are training diseases.

Key pairs. Pairs are mined within a split and reduced to a one-to-one matching, so that no question occurs in two pairs. Excluded: negatively phrased lead-ins, options that refer to other options, and questions without a case description. For training rows, ledgers are sampled once from the frozen rule-trained reader and kept if their quotations occur in the case and the frozen rule-trained judge returns the keyed verdict on all four cells (38.5% kept); the kept set is fixed and shared by all ledger formats. By tercile of vignette similarity, reversal of the critic is 55.7%, 44.1% and 33.8%. CareQA covers six subject areas; results by area are in the

repository, and the biology and chemistry areas contribute 4% of pairs.

CondMedQA. CondMedQA (Parekh et al., 2026) has 100 questions that contain a patient condition, each with the answer given the condition and the general answer that would hold without it; candidates were generated by a model, verified by its authors, and 30 were reviewed by annotators with medical expertise. We score both answers as claims for the question as posed and count how often the condition-specific answer is preferred: 41.0% for the critic, 56.0% for rule-only Ledger-RM [46, 66] and 80.0% for the closed judge. This is a test of resisting the default answer and not a reversal pair, since the benchmark has no version of the question without the condition; no rule is stated, so it also measures knowledge.

Claim-side applicability on real notes. EHRNote-ChatQA (Kim et al., 2026) asks questions about a patient’s discharge summaries, each with one correct and four incorrect answers that follow defined patterns and were reviewed by medical experts. Two patterns mirror our near-misses on the claim side: a correct fact attributed to the wrong admission, and a related entity or attribute taken from elsewhere in the notes. Scoring each answer as a claim against the notes, the correct answer is preferred to such a distractor in 58.0% of cases by the critic, in 69.5% by rule-only Ledger-RM and in 77.0% with clinical pairs. The data are under credentialed access and may not be sent to external services, so closed judges are not evaluated on them.

NLI4CT-P. Base macro-F1 is 71.0 for the critic and 76.4 for rule-only Ledger-RM; joint correctness over an original statement and all its interventions is 38.5% and 47.9%. Faithfulness and consistency follow the task definitions; pairs that share an original statement are resampled together.

Overlap. By normalised question text, CareQA shares no question with MedQA. The MedQA portion of ReMedQA (Cocchieri et al., 2026) lies within the MedQA test split; we do not use ReMedQA.

E Ledger Format and Scoring

need: penicillin allergy in the patient
found: "penicillin allergy (anaphylaxis)"
subject: other (mother)
status: present
time: current

Given this ledger, the rule and the claim “Order amoxicillin”, the judge returns +. Malformed ledgers occur in 0.4% of rule-tier and 2.4% of MedEinst instances. Temperature scaling for the combined reward is fitted on development data separately for the two scores. The verification pass prompts the same adapter with one ledger entry and the case and asks whether value, subject, status and time are as recorded; it rejects 41% of wrong entries and 3.2% of correct ones. In triplet and pair evaluations one ledger serves both claims. In selection the reader receives a step; for two candidate steps about the same criterion the ledgers are identical in 98.7% of cases. On L2 the reader is right on 96.9%, 95.4% and 94.1% of found, subject and time fields. In 2.1% of L2 cases it quotes a span that occurs in the case and is not the decisive finding; the string check cannot detect this.

F Diagnostics

Scorer (L2)	Rev	Hold	TA
Always default	0	100	0
Claim only	0	–	0
Concept named (h_k)	≤ 45	–	0
Attribute-blind, logistic	85.0	41.0	27.2
applicability triplets	93.0	11.5	6.3
Bag of words, case + claim	31.0	55.0	9.2
Trigger lexicon + program	88.0	58.5	52.0
with training cue phrases	97.0	93.5	91.0

Table 7: Shortcut scorers (%). Uncoloured entries follow from Proposition 1. Concept named: any deterministic scorer that uses the case only through h_k ; in 45% of triplets the flip introduces the decisive finding. Attribute-blind: a logistic model on concept–value features without subject, status and time. Trigger lexicon: a rule-based reader with negation, experiencer and temporality triggers in the style of ConText (Harkema et al., 2009), built from training cue phrases, followed by the rule program; its drop on held-out cue phrases is why test cues are held out.

Signal	Base	Ties	UR (n)	P	K	G
Inj.-error PRM	93.5	0.0	8.1 (4,769)	92.8	89.5	71.4
Qwen3-8B	80.2	0.0	24.6 (4,090)	95.3	91.2	56.8
Qwen3-32B	88.4	0.0	26.0 (4,508)	97.6	95.4	33.4
Claude Opus 5.5	96.0	3.8	12.9 (4,896)	99.2	98.1	11.3

Table 8: L0 (%). Base accuracy; tied preferences; unnecessary reversal among triplets with a correct base preference, with its denominator; reading and application reversal; composition gap.

Shift relative to a missing-input baseline. Let $d_0 = d(x^m)$, $\delta(x) = d(x) - d_0$, and $y \in \{+1, -1\}$ the correct sign of $d(x)$. For cases with $y d_0 < 0$, where the baseline preference opposes the label, $y d(x) = -|d_0| + y \delta(x)$, so the preference is right if and only if

$$y \delta(x) > |d_0|. \quad (3)$$

Cases with $d_0 = 0$ have no opposing baseline and are counted separately. We sort the others into four disjoint classes, in this order: *crossed* (Eq. 3); *unmoved* ($|\delta| \leq \tau$, the model’s median $|d(x^p) - d(x)|$ over presentation edits); *wrong direction* ($y\delta < 0$); *short*. The split restates the decision rule and adds the direction and relative size of the shift. The *shift ratio* compares that size with the model’s response to an error inside a step:

$$\kappa = \frac{\text{med}\{y \delta(x) : y d_0 < 0\}}{\text{med}\{u(s^+, x) - u(s^-, x)\}}, \quad (4)$$

where the denominator runs over matched pairs of a correct step s^+ and its injected-error version s^- on the same case in the MedPRMBench development split; both medians are in log-odds, and κ is reported with a bootstrap interval when the denominator is positive. For the injected-error PRM, 16.8% of baseline-conflicting cases cross, 55.2% move the right way and fall short, 19.5% do not move and 8.5% move the wrong way; $\kappa = 0.18$ (Figure 4).

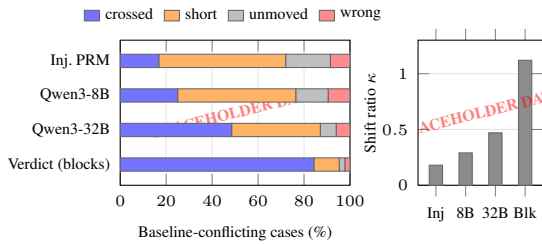


Figure 4: Rule tier. Left: cases whose correct preference opposes the missing-input baseline, by class (Eq. 3; models that expose log-probabilities). Right: shift ratio κ (Eq. 4). Verdict (blocks): the verdict-only model trained on flip blocks.

Readouts and sampling noise. Dedicated PRMs use their released scoring head, open judges the log-odds of the verdict token, and API judges without log-probabilities the verdict frequency over 32 samples at temperature 1.0, for which the standard error of a frequency is at most 0.09. Scoring Qwen3-32B by the same sampling protocol lowers its triplet accuracy from 45.3% to 42.0%, with 4.1% ties.

Input ablations. Replacing the question stem by “No case description is available” lowers PRM-Score on MedPRMBench by 1.2–2.9 points for trained PRMs. This is an input ablation and not a measure of grounding: many injected errors are detectable within the step, and chains restate patient facts. On the error types that the benchmark defines relative to the case the loss is 4.0–9.5 points, and 6.2–13.0 when numbers and quoted spans of the stem are also masked in the chain.

Exploratory. A probe on the residual stream of the critic predicts the criterion outcome on held-out signature classes with 90.8% accuracy (control task (Hewitt and Liang, 2019): 49.6%); decodability does not show use (Elazar et al., 2021).

G Additional Results

Variant (rule triplets only)	Rev	Hold	TA	MR	ME
Ledger-RM (facts only)	91.0	86.9	78.6	84.6	44.8
+ decision field	91.3	87.4	79.2	84.9	44.1
decision field only	90.8	86.0	77.5	83.0	38.2
program on predicted bit	90.9	86.2	77.8	83.1	—
reader writes bit only	80.5	74.0	63.0	79.0	34.5
+ verification pass	92.0	89.1	81.3	85.6	45.4
program-supplied ledger	98.6	97.8	96.5	97.0	—
One-stage ledger	88.1	80.2	71.6	82.3	47.5
Value ledger	90.4	76.4	68.3	83.5	42.0
Evidence summary	89.0	74.8	66.5	80.4	39.4
Concept scorer	84.5	78.0	65.9	76.2	31.0
Premise gate	62.0	79.5	49.3	70.1	30.5
– near-misses	90.2	49.0	44.0	83.9	46.3
+ probe re-weighting	84.0	60.5	50.8	83.0	44.0
– presentation edits	90.6	82.0	73.5	84.2	44.5
– ledger resampling	89.7	85.5	76.9	84.0	44.0
Verdict only, pairwise loss	77.0	71.2	53.6	60.5	33.8

Table 9: Ablations on L2 and MedEinst zero-shot (%). Decision field: the reader also writes whether the condition applies. Verification pass: every entry is re-read against the case and regenerated once if rejected, after Agand (2026). Program-supplied ledger: the judge receives the target ledger. Concept scorer: fields predicted by linear heads and a linear verdict. Premise gate: the step check, attenuated toward a neutral score when the finding a step relies on is not supported by a ledger extracted once per case, after Wang et al. (2026). Probe re-weighting: near-misses are not trained on; base-flip pairs whose preference shifts under near-miss and presentation edits are down-weighted, after Liu et al. (2026). Pairwise loss: Bradley–Terry on the two claims.

Training. LoRA rank 64, learning rate 1e-4, batch size 64, 2 epochs, $\lambda = 1.0$, $p = 0.3$. Rows with clinical pairs mix rule triplets and pairs 1:1.

System	L0	L1	L2	FR
Critic	27.9	26.3	25.1	5.3
Verdict only, triplets	79.3	64.0	52.4	5.6
Ledger-RM, rule triplets	92.4	85.5	78.6	5.2
Closed judge, ledger	86.7	85.9	85.1	5.4

this check is exploratory.

1290

Table 10: Rule ladder (triplet accuracy, %) and achieved false rejection on L2 at the development threshold. Rejection curves over thresholds and per-class results are in the repository; the lowest held-out class is at 66.0% for Ledger-RM.

Resources. Per scored claim: verdict only, 1 pass and 1 generated token; Ledger-RM, 2 passes and 45 tokens; generated-program verifier, 1 pass, 540 tokens and one execution; GenPRM-style verifier, 1 pass, 500 tokens and one execution; closed judge with ledger prompt, 1 API call and 210 tokens, about 60 times the list price of the 8B model per claim. Latency and hardware are in the repository.

Policy training. GRPO from one Qwen3-8B initialisation on rule triplets, 3 runs, 8 rollouts per prompt, 2000 steps, advantages standardised within a group. Before each run we check that reward terms are not collinear within groups and that the setup fits a 64-prompt subset. Accuracy on base-flip pairs of held-out structures, judged by the rule program, moves from 38.6% to 52.3% with an outcome reward, 33.9% with the injected-error PRM and 63.5% with Ledger-RM. A reference-graph reward after Mu et al. (2025), which scores a rollout by its coverage of the decisive findings and of the chain from finding to criterion to conclusion, gives 58.0%; the rule program supplies the graph, so this reward, like the outcome reward, uses the answer for each case.

Selection protocol. A trace is eligible if it contains a final answer (6.1% are not). Ranking by the product of the two calibrated scores gives 78.0% on MedQA and 66.1% on key pairs, a logistic combination fitted on development data 78.5% and 67.0%, and the minimum 78.3% and 66.7%. With a paraphrased vignette the combined reward gives 78.0% on MedQA. For nested pools the selected reward cannot fall as N grows, so we report selection regret, the gap between oracle and selected accuracy: at $N = 64$ it is 27.0 points for the step check and 15.5 for the combined reward on key pairs. On the open-ended version of CareQA, graded against reference answers by a fixed model grader, the share graded correct moves from 48.2% to 51.0%;